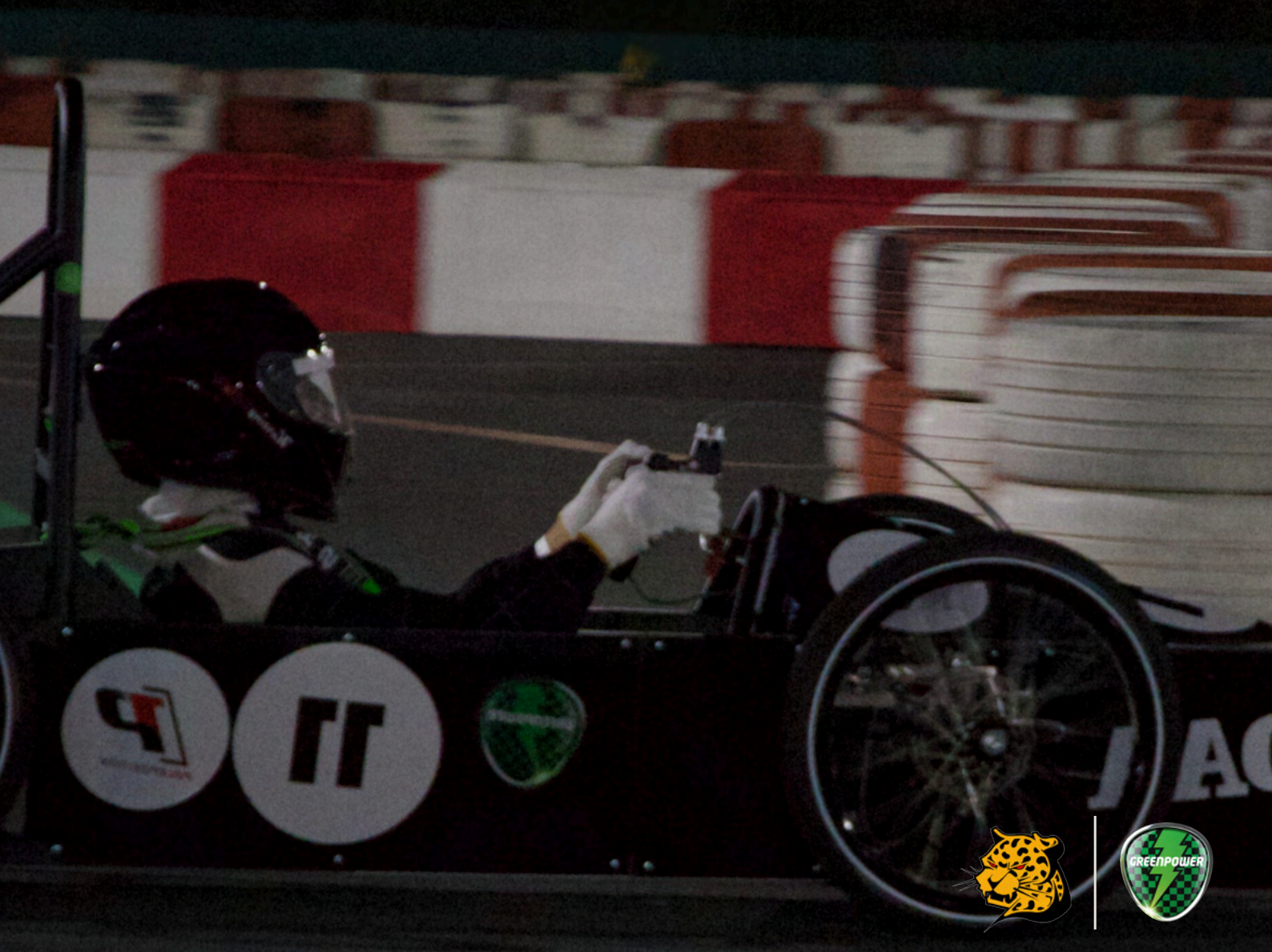


LEOPARDS PORTFOLIO 2025



RAISING THE BAR
EVERY YEAR



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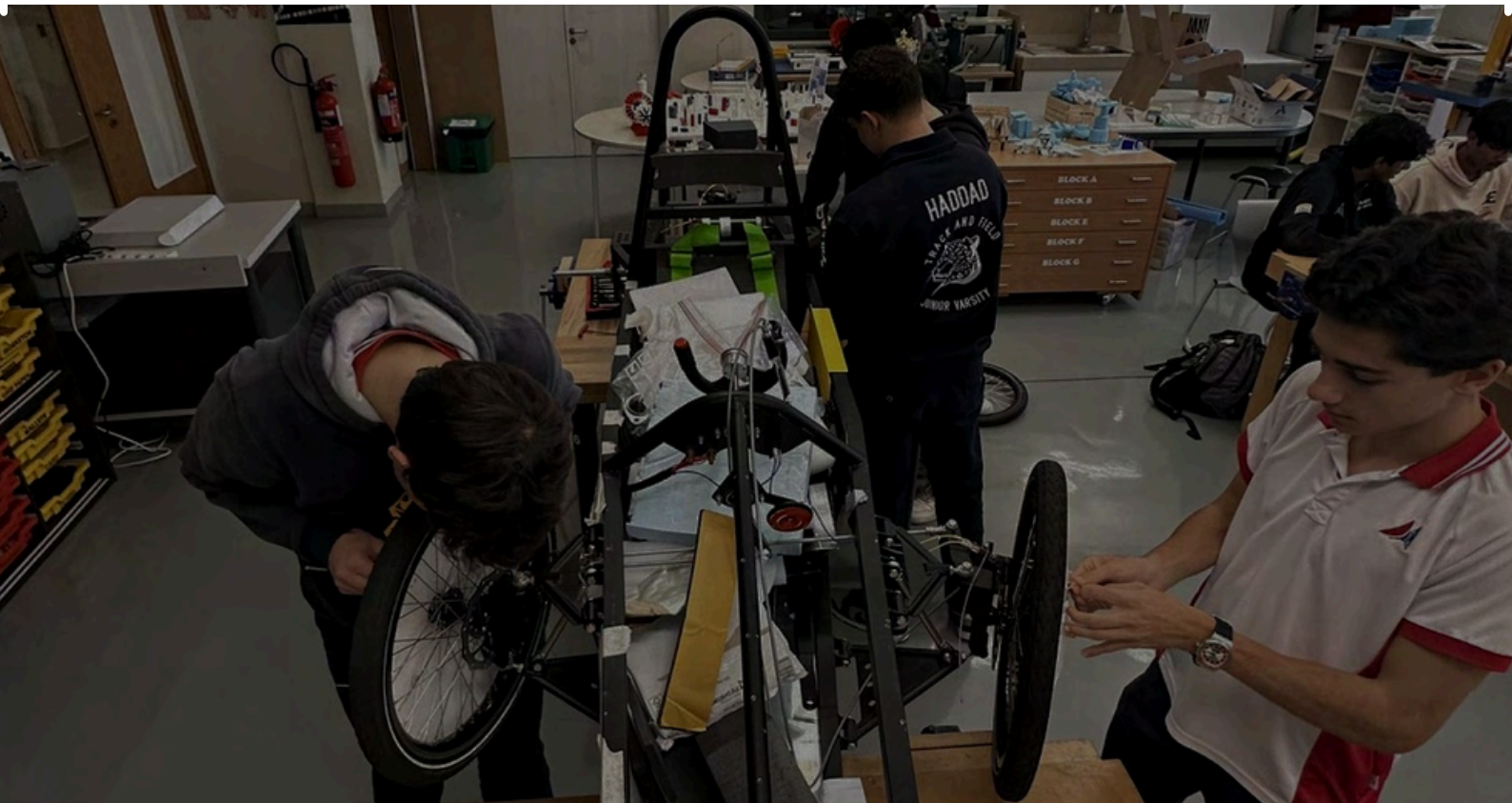


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ABOUT US

We are the Leopards, the reigning champions of GreenPower. We are a team of 12 students between grades 10 and 11 with a passion for engineering and innovation. By delegating our unique skills, we aim to curate an electric kart that exemplifies qualities of sustainability, innovation, and performance. From CAD to aerodynamics testing, we explored new applications and programs to develop new skills and hone our interest and passion in engineering. With our collective interest and passion for the sport, we are dedicated to pushing ourselves across the finish line with improvements from last year's competition as we constantly learn from our challenges. Whether it was minor upgrades like weight reduction or major upgrades like changing the entire nose cone, we were able to develop a car that will efficiently and quickly cross the finish line. We look forward to meeting new teams from across the nation with sportsmanship, integrity, and a fighting fire to win.

**WE ARE THE BEST
WE ARE ONE
WE ARE THE LEOPARDS
WATCH US ROAR!**

OUR TEAM

Team Supervisor
Padraig Gaughan



Sponsorship
Elanaïs Pichon, Ruhaan Sahi,
Eden Rebello, Niryan Parikh



Design Mechanics
Mauricio Krupka Buendia, Gio Haddad,
Fadi Kazak, Akbar Muratov,
Julian Decrauzat



Design Aesthetics
Ahaan Mehta, Rodrigo Vieira, Kareem
Aboulbrahim,



Design Aesthetics

Step 1: Brainstorm the ideas for new aesthetic pieces

Step 2: Sketch concepts out on paper with rough specifications

Step 3: Import sketches to Fusion360 and create a 3D model

Design Constraints & Considerations

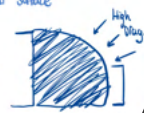
Our design followed strict specifications: the nose cone required 200mm of foam and a simple shape, while the car's length and width were capped to ensure fairness. We prioritized aerodynamics—testing nose cone shapes, using minimalist side panels for smooth airflow, and adding aero discs to reduce wheel drag. Lightweight aluminum was chosen for the side panels to balance sustainability and regulation-compliant weight. Sponsorship-related design challenges were minimal.

Entry 40

22/11/24

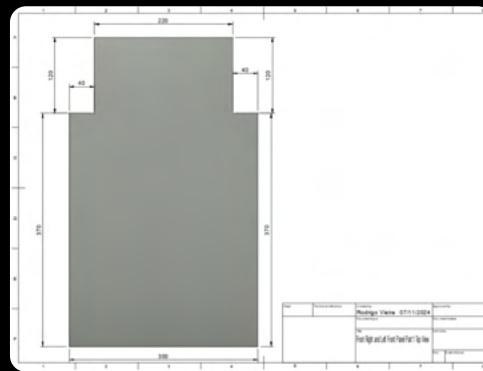
→ In the green power cube I'm going to be in charge of the aerodynamics. Initially we'll focus on the nose cone as that, alongside the driver, are the main source of air resistance.

→ Currently, we have a Quartercircle design that really does nothing for aerodynamics, as its perpendicular surface are to air flow is still very high. It also has a dent in the hood. This was caused by the Sun as they left it outside sitting during the race and the Sun heated the plastic to the melting point.



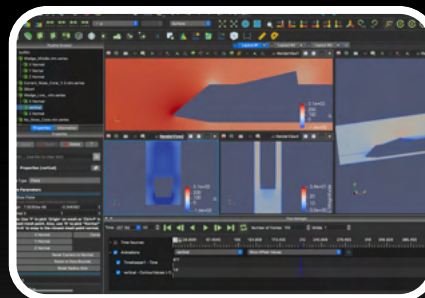
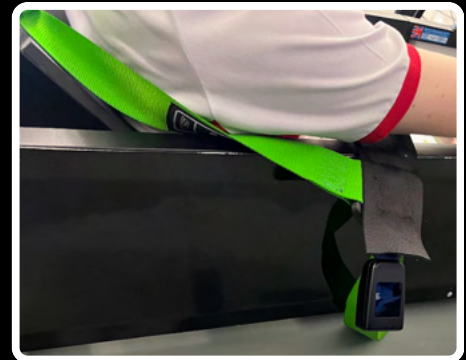
Sketch of the initial Nose Cone Design

Panel Laser Cutting Process



CAD Models were converted to orthographic drawings. These drawings went through a software called 2D Design

Laser-cut the prototypes. Ran paper prototypes before sending final designs for manufacturing.



Running aerodynamic simulations on the nose cones



The Final Product!

Design Mechanics

Main components of designing mechanics: Gear Ratios, Fixing brakes to pass strength tests, attaching the wheels and axles.

WE aimed to boost performance by adjusting the gear ratio—reducing the driven gear teeth from 72 to 60 for a higher top speed. We worked with the aesthetics team to redesign aero discs for consistent wheel attachment and used 3D-printed washers to align brake discs and reduce friction. We also calculated the correct chain length for the new gear setup and coordinated its adjustment with a mechanic.

Purpose of the Design Mechanics Team

Several challenges arose during the building process. Initial brake issues caused friction due to rubbing pads and discs; loosening them helped but weakened braking. This was resolved using a 3mm Allen key for fine adjustments and repositioning the brake light sensor. Some parts didn't fit smoothly into the axle, which was fixed using grease. Incorrect brake disc installation led to a full disassembly of the front wheels. In some cases, assembly steps had to be reversed—such as detaching axle bearings to access and install hidden screws properly. These experiences reinforced the importance of patience and precision over speed.



Final Frictionless
Brake disks



Initial (left) vs Final 3D
printed (right) washers



Before vs After teeth fix
on the drive gear



First iteration of aero
discs



Final iteration of aero
discs



Perfecting
chain length

SPONSORSHIPS

Our goal this year as a team was to raise and allocate funds to facilitate innovation and changes to our car.

Learning from last year, we decided to create a specific team focusing on branding and searching for sponsorships from the start of the season. This year, our team consisted of Niryan Parikh, Eden Rebello, Elanaïs Pichon, and Ruhaan Sahi.

The team's first project was to create a pitch to reach out to various companies for a sponsorship opportunity. This included 3 steps: a presentation, a video, and an email template.

Presentation: The goal of this step was to articulate what the Greenpower initiative is and what our role is and was in the competition. We outlined our previous year's performance and introduced our team. We also included different tiers that the potential sponsor could contribute through. This presentation was then sent out, along with the email template, to businesses and potential sponsors.



SPONSORSHIPS

Email Template: In order to reach a wider audience, our team curated a professional email template that we would use to cold email local and international businesses. We outlined the challenge, the team, and our goals before requesting them for a sponsorship and attaching the presentation. However, this method of outreach proved to be insignificant as larger companies have long bureaucratic systems in place for such actions, thus, it was not the most appropriate method.

Company Name	Contact	Member in charge	Status	Details of Agreement
DP World	simon.sonoo@dpworld.com	eden rebello	Awaiting Response	
Elden		eden rebello	Awaiting Response	
Deloitte	Mtomaszewicz@deloitte.com	Rodrigo Vieira	Rejected	
KPMG	gtraquina@kpmg.com	Rodrigo Vieira	Rejected	
Thales Group	ricardo.vieira@thalesgroup.com	Rodrigo Vieira	Awaiting Response	
Etisalat	gofernandes@eand.com	Rodrigo Vieira	Awaiting Response	
Continental	jose-luis.delafuente@conti.de	Rodrigo Vieira	Rejected	
ZEEKR	Contact given by continental	Rodrigo Vieira	Awaiting Response	
Virgin Radio				
scaffco	omar@scaffco.com	Kareem AbouIbrahim	Agreement reached	Materials fully covered excluding nose con
Standard Charter		Elañais Pichon		
Gatorade		Elañais Pichon	Awaiting Response	
Pepsi		Elañais Pichon	Awaiting Response	
Visa		Elañais Pichon	Awaiting Response	
Colas		Elañais Pichon	Awaiting Response	
Puma		Elañais Pichon	Awaiting Response	
Under Armour		Elañais Pichon	Awaiting Response	
Adidas		Elañais Pichon	Awaiting Response	
Google		Elañais Pichon	Awaiting Response	
NAX Holdings		eden rebello	Rejected	

You are viewing this template as my previous sponsor. Please click on the "Sponsorship Request" button to view the full template and request form. It includes information on how to use the form.

Sponsorship Request: Dubai American Academy's Greenpower Challenge Team - [Name]

Dear [Company/Person],

I hope this email finds you well. My name is [Name], and I am a student at Dubai American Academy (DAA), where I am currently participating in an exciting and educational project called the Greenpower Challenge, in which we design an electric car and engineer it to be the fastest around the track. This initiative is designed to inspire and educate students in the fields of engineering, teamwork, sustainable energy, and innovation.

Our project involves building a kart using a frame and resources provided by the challenge organizers. The core objective is to create a vehicle that not only functions efficiently but also adheres to sustainable engineering principles. Additionally, we must manufacture unique body panels that are designed by our team. Last May, our team successfully curated the fastest kart on track along with a successful race strategy, thus, winning the first place prize in the competition. This year, we aim to produce the most slipstream kart, thus, making it faster. This hands-on experience is a fantastic opportunity for us to apply theoretical knowledge in a practical setting, fostering skills that are crucial for future engineers. Here are some posts and an article that we were featured in: [Post #1](#), [Post #2](#), [Article](#)

However, there is one significant challenge that we face: the requirement of funding to produce our body panels and purchase the necessary accessories for the competition. These panels and resources are crucial for the completion of our project, and unfortunately, they are not included in the equipment provided by the organizers. This is where we hope your esteemed organization can play a pivotal role.

We are reaching out to seek sponsorship to acquire the necessary funding. Your support would not only help us complete our project but also contribute to promoting sustainable engineering practices among young students. Additionally, this sponsorship promotes the transition to electric vehicles leading to a sustainable world. By sponsoring us, [Company] will be directly impacting the education and development of future innovators and environmental stewards in return for the following benefits for different prices (can be negotiated):

- Silver: A Social Media Post* (Video) + Product Placement (500 AED)
- Gold: Silver + Logo on Uniform + Rights of social media post(s)* (1500 AED)
- Platinum: Gold + Logo on car + Multiple Social Media Posts* (Video) (3500 AED)
- Diamond: Platinum + Team Name (Highest Sponsor - 5000+ AED)

*Social Media post on official DAA account (15.2k followers)

We believe that this partnership can be mutually beneficial for both parties. You will be able to promote your brand or product while helping us achieve our goal and win the GreenPower Challenge again this year. We have attached a sponsorship pitch deck with a few more details about this proposition.

We would be thrilled to discuss this opportunity further and answer any questions you may have. Your consideration of our request means a great deal to us, and we are hopeful for a positive response. If it would be preferable to you, we are willing to come to your company and pitch our proposition.

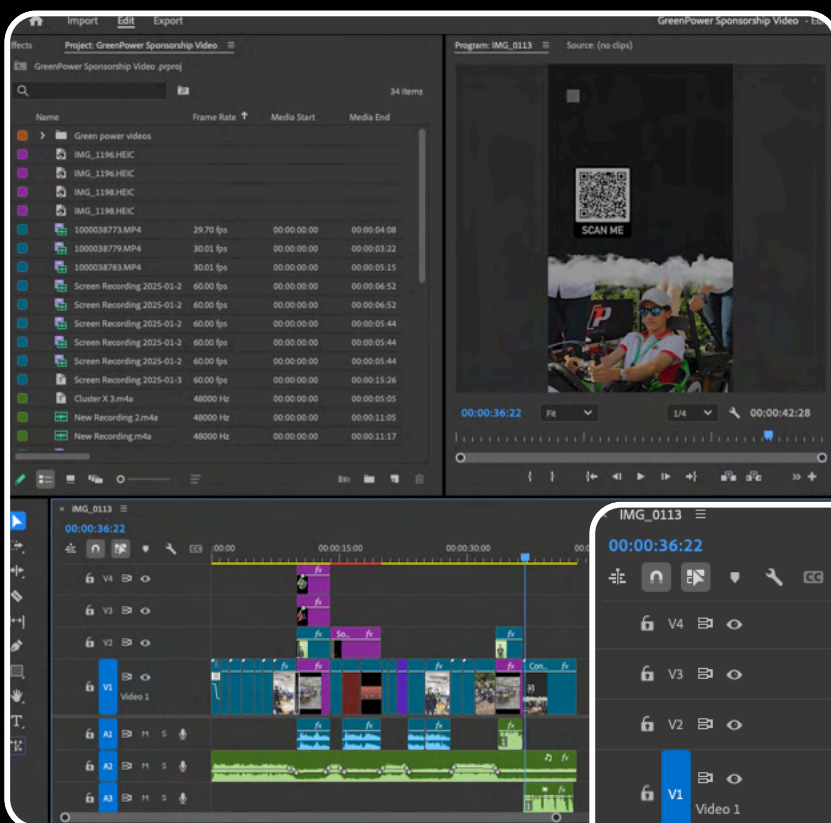
Thank you very much for considering our request. We look forward to the possibility of partnering with [Company] in this exciting endeavor.

Best regards,

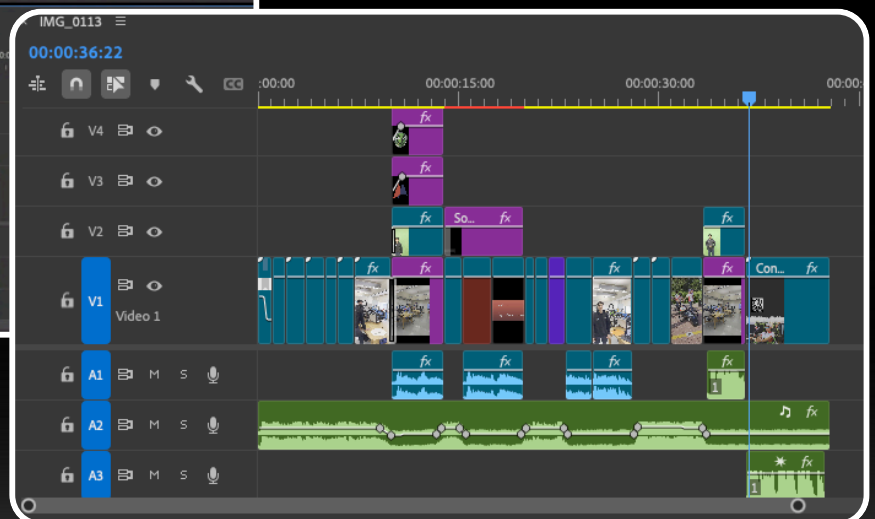
[Name] and DAA Greenpower Challenge Group
[School email@gemsdaa.net]
Student, Dubai American Academy
Coordinator
Adraig Gaughan
[School email@gemsdaa.net]

SPONSORSHIPS

Video: Another way our team decided to conduct outreach was to create a minute-long reel and upload it to our school's Instagram page to find willing parents who are interested in sponsoring the innovative initiative this year. We created scripts for the team leaders to read, outlining what they do and their role in the Leopards team this year, filmed their respective parts and videos of the team hard at work, and finally edited the short video together to create a cohesive outreach video. Despite an ideal final product, we were unable to upload the video as part of the school, as it requests money from parents of students who may attend DAA.



Understanding and cooperating with the school's corporate policies and systems, we had to consider a different approach. Our connections.



OUR SPONSORS

SCAFFCO: One member of our Greenpower team had strong ties with a scaffolding company in Dubai named SCAFFCO. Utilizing this relationship, we were able to approach them with a sponsorship offer. They produce our body panels in exchange for marketing their brand on our car and uniform.

DAAPA: To work around the inability to reach sources of funding from external parties, we decided to resort to in-house resources such as the DAA Parent Association, which encourages student organizations and initiatives. Through another member of our team and their personal relationship with the DAAPA we were able to get in contact. Using our materials that we created before hand, we pitched the Greenpower challenge to them and what the Leopards team aspired to be. After some discussion we were able to finally allocate funds for this year's kart.

Expense Breakdown:

Item	Quantity	Links	Total (AED)
DAAPA	-	-	5000
Helmet	2	Helmet	(858)
Race Suits	3	Race Suits	(531)
Gloves	3	Gloves	(944)
Team T-shirts	14	-	(698.25)
Transport	-	-	(≈1000)
Merchandise	40	-	(≈800)
Total	-	-	168.25

SUSTAINABILITY AND GOALS

As the DAA team, we exemplify ourselves as being exceptionally sustainable, a crucial concept during this Greenpower project. Before commencing our kart creation process, we established two ground rules.

The first rule was to work digitally the entire time. This meant staying away from using paper, which will unlikely be put to use after writing on it. Before the 3d modelling on Fusion 360, team members opted to sketch the potential kart design on an iPad; a reliable technology that allows for constant improvements and upgrades, without costing the life of another tree.

Another rule we managed to keep committed to was the no use of plastic water bottles. This process was hard work, and drinking water was important, especially in the UAE heat. For this reason, each member carried their reusable water bottle, ensuring a plastic-free experience.

While building the car, we would receive pieces here and there for helmets and parts, giving us a range of plastic wrappers. Being oriented with a sustainable mind, our team was dedicated to recycling the wrappers into the correct bins placed outside of the workspace, so that we had a low waste approach.

Having built the kart last year, our team was able to recycle the side panels, using durable aluminium as the primary material. Aluminium is 100% recyclable and lightweight, making it easy to transport, reducing carbon emissions. Alongside this, our team used the same foam for the seat and interior protection, eliminating the need to get new foam, wasting more material.



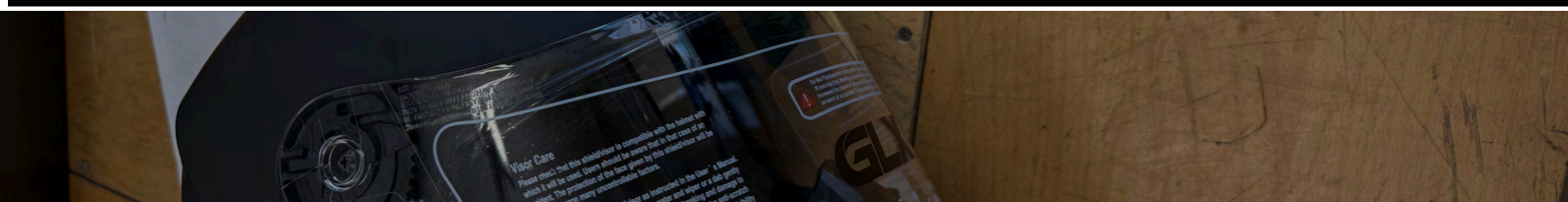
As we approached the race, we started to think about how we could make our journey come to an end in a sustainable finishing touch. Therefore, we decided to take a bus, collectively the day of the race, limiting the amount of fuel released from the drive to Abu Dhabi.

Reflection

As a team, participating in this Greenpower challenge was an incredible experience. We had the chance to grow as not only learners, but also as people. From the very beginning, we were determined to make a go-kart that not only was the fastest, giving us a chance to win, but also to do it using sustainable considerations. Throughout the process, we completed steps that would help progress our kart while being eco-friendly. The acts of recycling wrappers, or using recyclable water bottles, the whole time we worked, are what help define us as a team; a committed community, wanting to win together. We tried to use items that have been previously used, like the foam and the side panels. We also learned a lot about teamwork, problem-solving, and the importance of being organized. We learned quickly that without a plan, our ideas could not come together.



Every challenge that appeared, we took on as a team, thinking critically and creatively. By the end of this project, we will not only be walking away with a successfully built, sustainable go-kart, but also a greater mind. Thinking that understands how participating in small actions can turn into solving larger goals. Together, we proved that sustainable engineering is possible, and many of us will be closing this chapter for the year with a passion for something incredible new.



The Leopard Collage

